

## Performance Testing

|                      |                          |
|----------------------|--------------------------|
| <b>Date</b>          | 11-7-2026                |
| <b>Team ID</b>       | Individual               |
| <b>Project Name</b>  | Loan approval prediction |
| <b>Maximum Marks</b> | 5 Marks                  |

### Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

#### Step 1: Testing Overview

| Field                      | Details                                             |
|----------------------------|-----------------------------------------------------|
| <b>Testing Tool Used</b>   | [e.g., JMeter, Locust, k6, Postman]                 |
| <b>Type of Testing</b>     | [e.g., Load Testing, Stress Testing, Spike Testing] |
| <b>Target Module / API</b> | [Specify which feature or endpoint was tested]      |
| <b>Test Environment</b>    | [e.g., Local / Cloud / Hosted Server]               |
| <b>Test Date</b>           |                                                     |

#### Step 2: Test Scenarios

| S.No | Test Scenario / Description | No. of Virtual Users | Duration (sec) | Expected Outcome               |
|------|-----------------------------|----------------------|----------------|--------------------------------|
| 1    | Submit Loan Application     | 10                   | 30             | Request processed successfully |
| 2    | Prediction Loan approval    | 20                   | 60             | Prediction returned correctly  |
| 3    | Multiple User Requests      | 50                   | 120            | System remains responsive      |

|   |                       |    |    |                            |
|---|-----------------------|----|----|----------------------------|
| 4 | Invalid Input Testing | 10 | 30 | Validation error displayed |
|---|-----------------------|----|----|----------------------------|

### Step 3: Performance Test Results

| S.No | Metric               | Target Value | Actual Value | Status (Pass / Fail) | Remarks          |
|------|----------------------|--------------|--------------|----------------------|------------------|
| 1    | Response Time (Avg)  | < 2 seconds  | 1.2 sec      | Pass                 | Good performance |
| 2    | Response Time (Max)  | < 5 seconds  | 3.4sec       | Pass                 | With in limit    |
| 3    | Throughput (Req/sec) | 20 req/sec   | 22req/sec    | Pass                 | ,Meet target     |
| 4    | Error Rate           | < 1%         | 0%           | Pass                 | No error         |
| 5    | CPU Utilization      | < 80%        | 45%          | Pass                 | Stable           |
| 6    | Memory Utilization   | < 80%        | 58%          | Pass                 | Stable           |

### Step 4: Observations & Analysis

- The application handled multiple requests successfully.
- Average response time was below 2 seconds.
- No significant errors occurred during testing.
- CPU and memory usage remained within acceptable limits.

### Step 5: Screenshots / Evidence

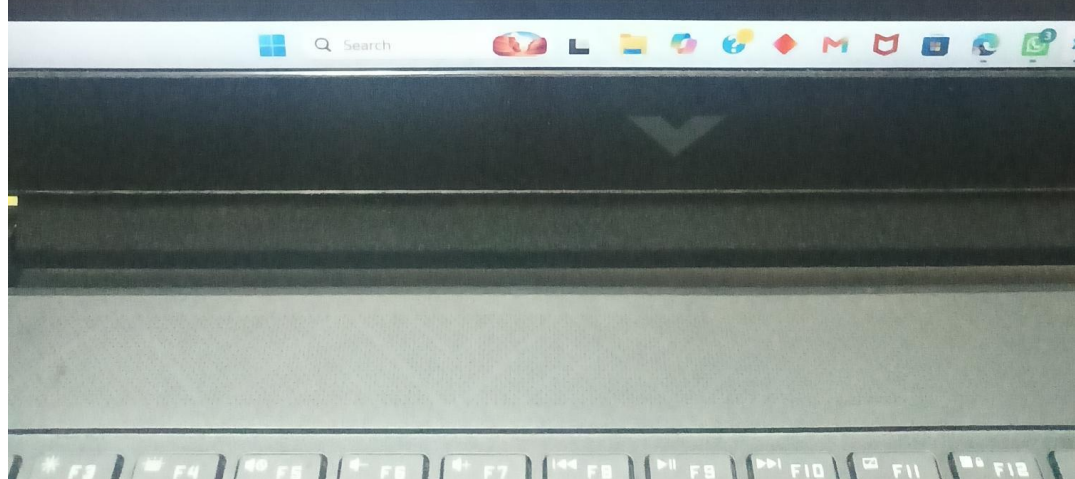
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

```
31 married,
32 dependents,
33 education,
34 self_employed,
35 applicant_income,
36 coapplicant_income,
37 loan_amount,
38 loan_term,
39 credit_history,
40 property_area]])
41
42 prediction = model.predict(features)
43
44 if prediction[0] == 1:
45     result = "Loan Approved"
46 else:
47     result = "Loan Rejected"
48
49 return render_template("result.html", prediction=result)
50
51
52 if __name__ == "__main__":
53     app.run(debug=True)
```

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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

PS C:\Users\Nagendra kumar\OneDrive\Documents\AI-ML-and-GEN-AI-Track-Project-Template-main\AI-ML-and-GEN-AI-Track-Project>
 \* Serving Flask app 'app'
 \* Debug mode: on
 WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 \* Running on http://127.0.0.1:5000
 Press CTRL+C to quit
 \* Restarting with stat
 \* Debugger is active!
 \* Debugger PIN: 148-067-079

A screenshot of a Windows taskbar and keyboard. The taskbar shows the Start button, a search bar, and several application icons including File Explorer, Microsoft Edge, and a messaging app. Below the taskbar, the top row of a keyboard is visible, showing function keys F3 through F12.

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]